

Euclid Book I — Reconstruction

CGJteamLab

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Chapter 1

Proposition I.1

Euclid's first proposition in Book I is usually presented as the straightforward construction obtained from the intersection of two circles. A formal reconstruction, however, reveals two logically independent problems: the existence of a common point of the corresponding circles and the proof that the resulting point does not lie on the line determined by the given base.

The proof of non-collinearity relies on the classical idea that a proper part of a segment cannot be congruent to the entire segment. This argument is well known from modern axiomatic reconstructions of geometry, including the work of Michael Beeson and his collaborators.

The purpose of this note is not to present a new geometric proof. Rather, it is to demonstrate that, once Book Zero has been developed, this argument becomes a direct application of the previously established theory. Proposition I.1 therefore serves as the first experimental test of Book Zero as an intermediate language for synthetic proofs.

1.1 The Classical Construction

Let $A \neq B$. Euclid constructs two circles of radius AB : the first centered at A , the second centered at B . One of their points of intersection is denoted by C .

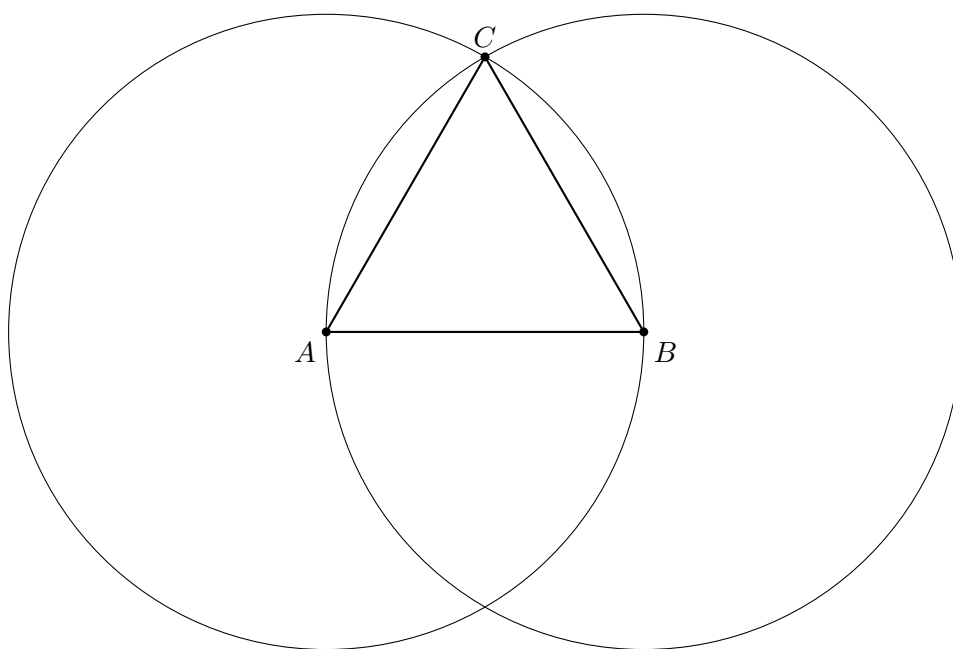


Figure 1.1: Euclid's classical construction. The point C is a common point of two circles of radius AB .

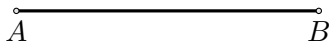
From the construction we obtain

$$AC \cong AB, \quad BC \cong AB.$$

Formally, however, this is not yet a complete proof. One must also show that the points A, B, C are not collinear.

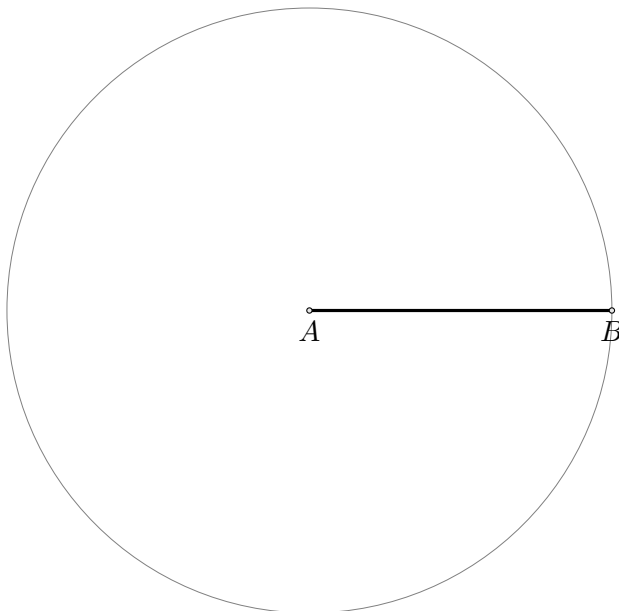
Step 1. Given Segment

The construction starts with a given segment AB .



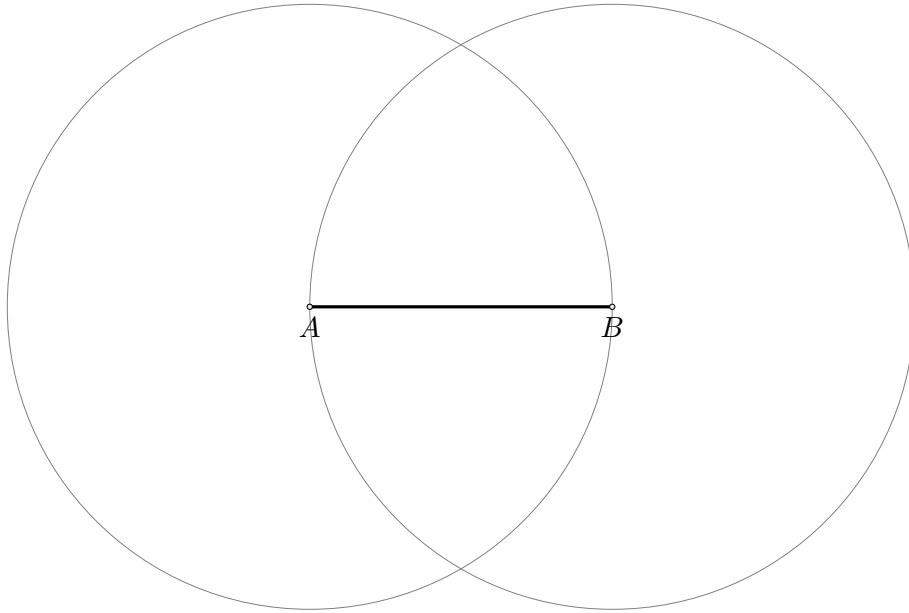
Step 2. Draw the Circle Centered at A

Draw the circle centered at A through B .



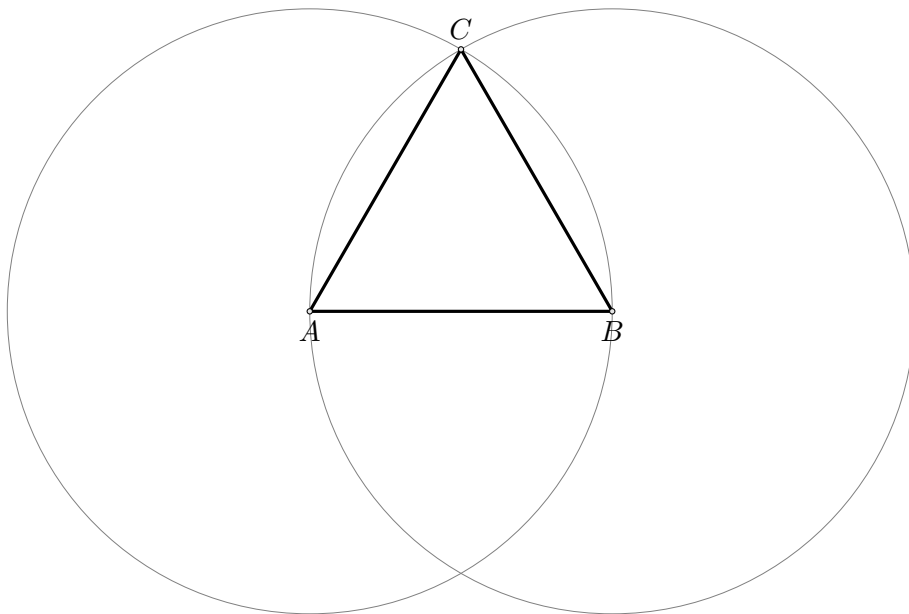
Step 3. Draw the Circle Centered at B

Draw the circle centered at B through A .



Step 4. Construct the Equilateral Triangle

Let C be one of the intersection points of the two circles. Join AC and BC to obtain the equilateral triangle.



1.2 Separating the Two Problems

The reconstruction reveals two logically independent stages.

1. **Existence stage.** One must obtain a point C such that

$$AC \cong AB, \quad BC \cong AB.$$

In the current version of the project, this stage is provided by the temporary principle `hilbert_equidistant_point_exists`. Ultimately, it is intended to be derived from an appropriate theory of continuity in Hilbert geometry.

2. **Synthetic stage.** From the congruences above one must derive

$$\neg \text{Collinear}(A, B, C).$$

This stage no longer requires any theory of circles.

1.3 Theorem

Theorem. Let $A \neq B$. Assume that there exists a point C such that

$$AC \cong AB, \quad BC \cong AB.$$

Then the points A, B, C are not collinear.

1.4 Proof

First observe that $C \neq A$ and $C \neq B$. If $C = A$, then $AC \cong AB$ would imply that a null segment is congruent to the non-null segment AB . Similarly, $C \neq B$.

Now suppose, for contradiction, that the points A, B, C are collinear. Since they are pairwise distinct, the trichotomy of the order of points on a line gives exactly three possibilities:

$$A - B - C, \quad B - A - C, \quad A - C - B.$$

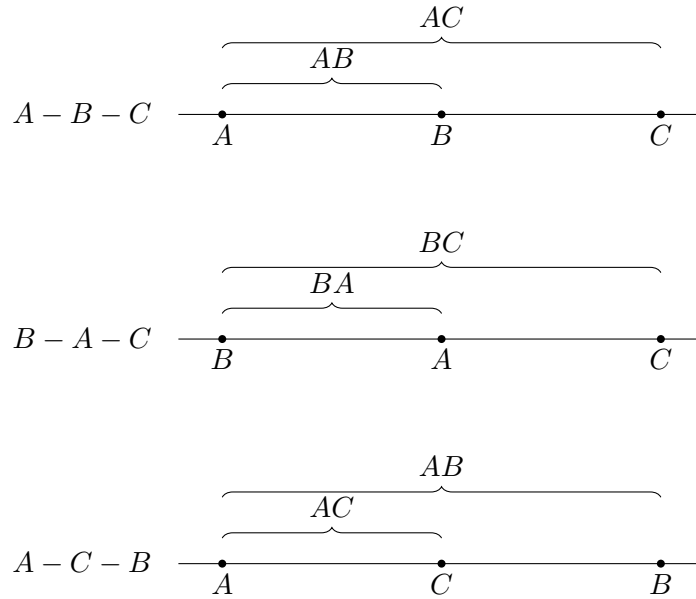


Figure 1.2: The three possible arrangements of pairwise distinct collinear points.

Case 1: $A - B - C$

The segment AB is a proper part of the segment AC . Book Zero contains the general theorem:

A proper part of a segment cannot be congruent to the whole segment.

On the other hand, by assumption,

$$AC \cong AB.$$

Using symmetry of congruence, we obtain

$$AB \cong AC,$$

which is a contradiction.

Case 2: $B - A - C$

The segment BA is a proper part of the segment BC . Since

$$BC \cong AB,$$

and reversing the endpoints of a segment does not change its length, we have

$$BA \cong AB.$$

By transitivity of congruence, it follows that

$$BA \cong BC,$$

again contradicting the fact that a proper part cannot be congruent to the whole.

Case 3: $A - C - B$

The segment AC is a proper part of the segment AB . But the assumption directly gives

$$AC \cong AB.$$

This yields the third contradiction.

All possible collinear arrangements lead to a contradiction. Therefore,

$$\neg \text{Collinear}(A, B, C).$$

□

1.5 Architecture of the Proof

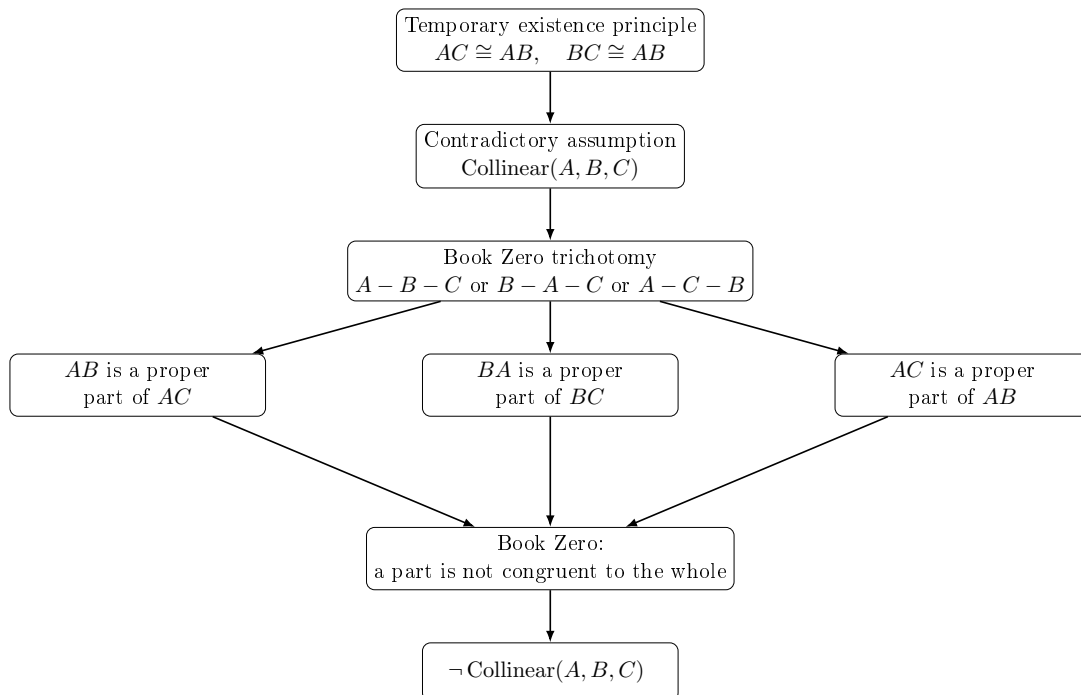


Figure 1.3: The structure of the proof of Proposition I.1 above the Book Zero layer.

1.6 Significance of the Reconstruction

The geometric result itself is not new. Constructing an equilateral triangle on a given segment is one of the classical theorems of elementary geometry.

Likewise, the idea underlying the proof of non-collinearity is not new. Modern axiomatic reconstructions, in particular the work of Michael Beeson and his collaborators, employ an analogous argument based on the theorem that a proper part of a segment cannot be congruent to the entire segment.

The novelty of the present reconstruction lies elsewhere.

The objective of the CGJteamLab project was not to discover a new proof of Proposition I.1, but to investigate whether the theory developed in Book Zero is sufficiently rich to serve as an intermediate language for synthetic proofs.

The experiment shows that, once the existence problem for an equidistant point has been isolated, the entire argument can be expressed almost exclusively in terms of the theorems previously established in Book Zero.

From the perspective of the architecture of the theory, this is more significant than Proposition I.1 itself. It demonstrates that Book Zero is not merely a collection of auxiliary lemmas, but rather forms a new layer of abstraction between Hilbert's axioms and classical synthetic reasoning.

Schematically, the situation may be represented as

$$\text{Hilbert's axioms} \longrightarrow \text{Book Zero} \longrightarrow \text{Proposition I.1.}$$

In this sense, Proposition I.1 constitutes the first experimental test confirming that Book Zero can serve as a language of synthetic proofs for the further reconstruction of Euclidean geometry.

Chapter 2

Proposition I.2

This note compares Euclid's original construction for Proposition I.2 with its reconstruction over the Hilbert foundation. In the CGJteamLab development the proposition follows immediately from Book Zero #49 (Layoff), illustrating the difference between the architectural roles of segment construction in Euclid's geometry and the Hilbert axiomatic framework.

2.1 Statement

Given a point A and a segment BC , construct a point D such that

$$AD \cong BC.$$

2.2 Euclid's Classical Construction

Euclid's construction proceeds as follows.

1. Construct the equilateral triangle DAB on the segment AB .
2. Produce the side DB beyond B .
3. Produce the side DA beyond A .
4. Draw the circle centered at B with radius BC .
5. Draw the circle centered at D with radius DG .
6. The constructed point L satisfies

$$AL \cong BC.$$

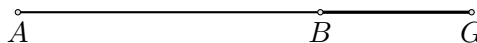
Step 1. Given data

The construction starts with a given point A and a given segment BG .



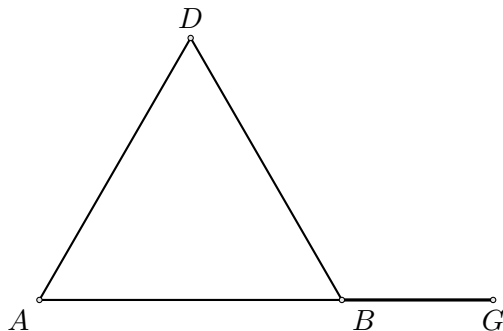
Step 2. Join AB

Join the given point A to the endpoint B of the given segment.

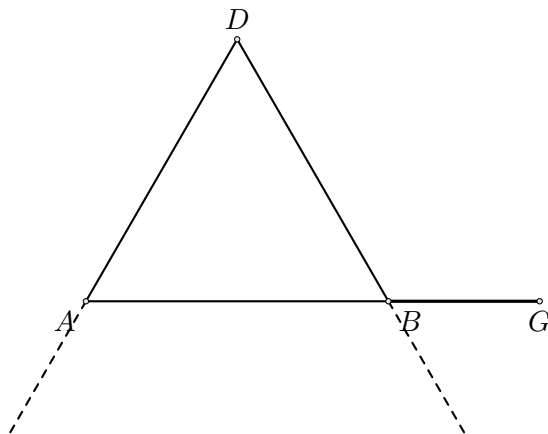


Step 3. Construct the equilateral triangle DAB

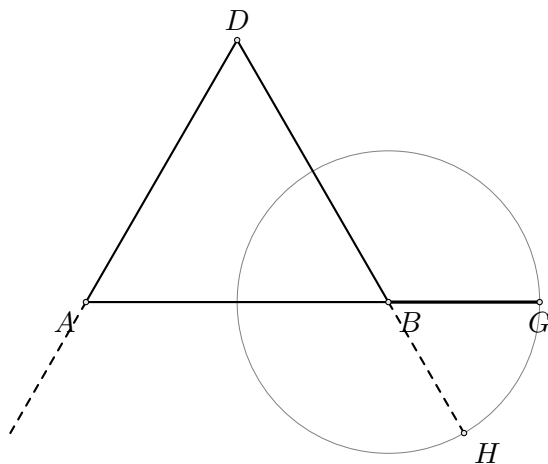
Construct the equilateral triangle DAB on the segment AB .

**Step 4. Produce DA and DB**

Produce DA beyond A and DB beyond B .

**Step 5. Draw the circle centered at B**

Draw the circle centered at B through G . Let H be the intersection of this circle with the line DB on the side beyond B .

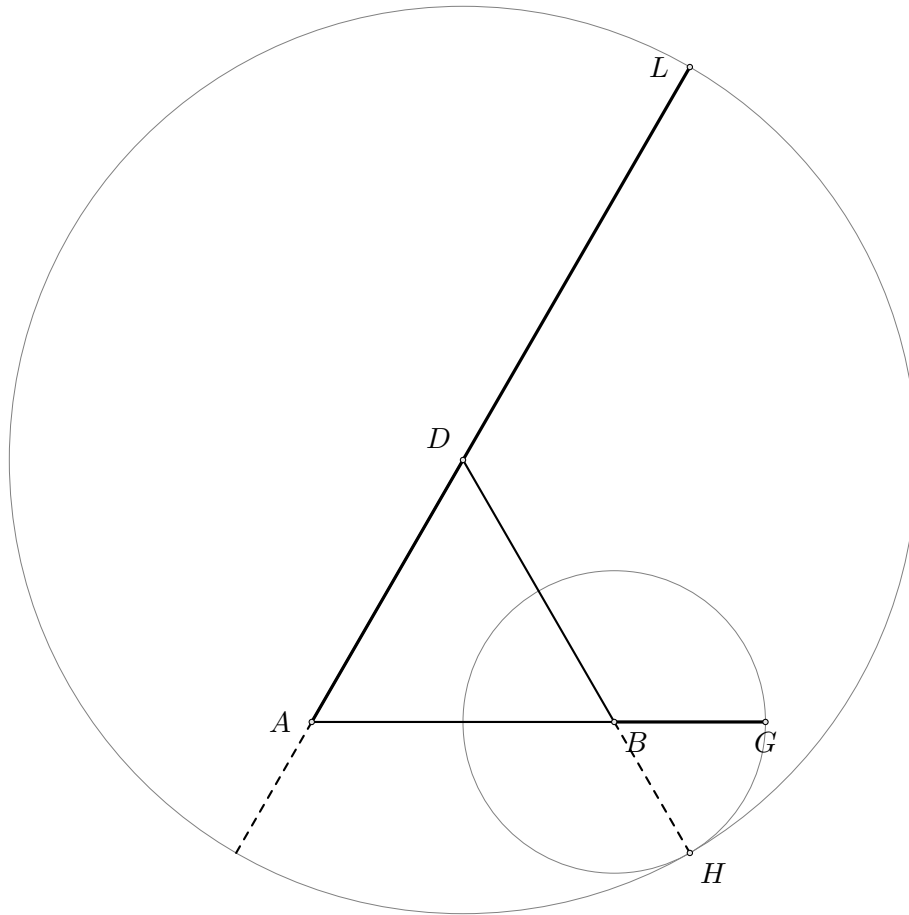


Since H lies on the circle centered at B through G ,

$$BH \cong BG.$$

Step 6. Draw the circle centered at D

Draw the circle centered at D through H . Let L be the intersection of this circle with the line DA on the side beyond A .



Since L lies on the circle centered at D through H ,

$$DL \cong DH.$$

The segment AL is therefore the required copy of the given segment.

2.3 Hilbert Reconstruction

The reconstruction over the Hilbert foundation is considerably shorter.

1. Choose any point $E \neq A$.
2. The ray AE is thereby determined.
3. Apply Book Zero #49 (Layoff).
4. Obtain a point D on the ray AE satisfying

$$AD \cong BC.$$

No auxiliary geometric construction is required.

Book Zero #49 provides a point D on a chosen ray from A such that

$$AD \cong BC.$$



Figure 2.1: Lay off the segment on a chosen ray.

2.4 Formalization

The formal proof consists of a single application of Book Zero #49.

```
theorem euclid_proposition_2 ...
  := bookZero_49_layoff ...
```

2.5 Discussion

Thus Proposition I.2 illustrates a fundamental architectural difference between Euclid's geometry and the Hilbert approach.

In Euclid's *Elements*, the transport of a segment is obtained through a sequence of explicit geometric constructions beginning with Proposition I.1.

In contrast, Hilbert incorporates this operation into the congruence axioms. Book Zero encapsulates it in the Layoff Theorem, making Proposition I.2 an immediate consequence.

2.6 Book Zero #49: Layoff

The proof of Proposition I.2 relies on one of the elementary construction theorems established in Book Zero.

Book Zero #49 (Layoff). Let $A \neq B$ and $C \neq D$. Then there exists a point X on the ray AB such that

$$AX \cong CD.$$

In other words, every nondegenerate segment can be copied onto an arbitrarily chosen ray while preserving its length.

Input

Ray AB

A----->

Segment CD

C-----D

|

| Book Zero #49

v

Output

A-----X----->

AX congruent CD

This theorem provides the fundamental operation of transporting segments. Once it is available, Euclid's Proposition I.2 becomes an immediate consequence.

Chapter 3

Proposition I.3

3.1 Introduction

Euclid's third proposition asks for the construction of a segment equal to a given shorter segment on a given greater segment.

Classically, this construction is obtained by combining Proposition I.2 with an additional circle construction. A segment equal to the shorter one is first laid off from an endpoint of the greater segment, and a circle is then used to determine the required point on the greater segment.

In the present reconstruction over Hilbert geometry, the logical structure is substantially different. The theory developed in Book Zero already contains an abstract notion of one segment being less than another. This notion is defined by the existence of a point between the endpoints of the greater segment such that the corresponding subsegment is congruent to the smaller one.

Consequently, the existential content of Euclid's Proposition I.3 is already embodied in the definition of segment inequality. The reconstruction therefore consists not in repeating Euclid's geometric construction, but in exposing the witness contained in the definition of `HilbertSegmentLess`.

This proposition provides the first clear example where a classical Euclidean construction becomes an immediate consequence of the abstract language developed in Book Zero.

3.2 The Classical Construction

Euclid's third proposition asks for the construction of a segment equal to a given shorter segment on a given greater segment.

Unlike Proposition I.2, the present construction does not introduce a new geometric technique. Instead, it combines Proposition I.2 with one additional circle construction.

The construction proceeds as follows.

3.2.1 Final Configuration

The objective is to determine a point E on the greater segment AB such that

$$AE \cong CD.$$

The final geometric configuration is shown in Figure 1.

From the construction we obtain

$$AE \cong CD.$$

The remaining task is to justify that the point E lies between the endpoints of the greater segment.

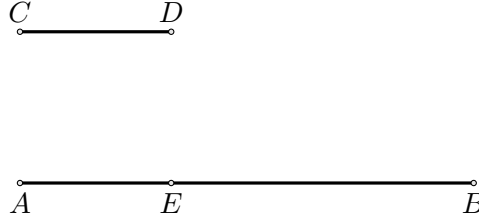


Figure 3.1: Final configuration of Euclid's construction for Proposition I.3.

3.2.2 Construction Algorithm

The construction uses Proposition I.2 as an already established construction primitive.

Step 1. Given Segments

Let AB be the greater segment and CD the given shorter segment.

Step 2. Apply Proposition I.2

By Proposition I.2, construct a segment AF congruent to the given segment CD .

Thus

$$AF \cong CD.$$

Step 3. Draw the Circle Centered at A

Draw the circle centered at A passing through F .

Step 4. Determine the Required Point

Let E be the intersection of the circle with the greater segment.

Since

$$AE \cong AF$$

and

$$AF \cong CD,$$

it follows that

$$AE \cong CD.$$

The required segment has therefore been obtained.

3.3 Proof

The classical Euclidean proof constructs the required point by applying Proposition I.2 and then introducing an auxiliary circle.

The reconstruction over Hilbert geometry follows a different logical route. The relation that one segment is less than another is already represented in Book Zero by the predicate

$$\text{HilbertSegmentLess } CD \ AB.$$

By definition, this predicate means that there exists a point E between the endpoints of the greater segment satisfying

$$A - E - B$$

and

$$AE \cong CD.$$

Therefore, assuming

$$CD < AB,$$

we immediately obtain a point E such that

$$A - E - B \quad \text{and} \quad AE \cong CD.$$

This is precisely the conclusion required by Proposition I.3.

Consequently, over the present Hilbert/Book Zero layer, the proposition is an immediate consequence of the definition of `HilbertSegmentLess`. □

3.4 Proof Architecture

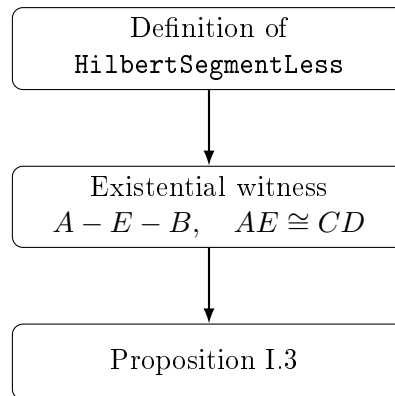


Figure 3.2: Logical structure of Proposition I.3 over the Book Zero layer.

3.5 Hilbert Reconstruction

Unlike Euclid's original proof, the present reconstruction does not repeat the geometric construction based on Proposition I.2.

The abstract notion of one segment being less than another has already been introduced in Book Zero by the predicate `HilbertSegmentLess`. By definition, this predicate contains exactly the existential information required by Proposition I.3.

Consequently, the reconstruction consists simply in extracting the witness from this definition.

This illustrates an important feature of the Book Zero layer. Classical geometric constructions are progressively replaced by abstract synthetic notions whose mathematical content has already been established.

3.6 Lean Formalization

The Lean formalization closely follows the mathematical argument.

The hypothesis

`HilbertSegmentLess` $CD\ AB$

already contains an existential witness providing the required point between the endpoints of the greater segment together with the corresponding congruence.

The formal proof therefore consists only of unpacking this witness and applying the symmetry of segment congruence.

Unlike the classical Euclidean proof, no additional geometric construction is required.

The resulting Lean theorem is therefore considerably shorter than the corresponding classical argument.

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